

DepoIndex

AI-Powered Deposition Topic Index

Automated, provenance-preserving topic indexing for deposition transcripts – built for litigation teams who need fast, defensible navigation of long testimony.

Problem

Long transcripts wander across subjects, repeat topics, and include brief digressions that make manual indexing slow, inconsistent, and error-prone.

Solution

Automatically detect meaningful topics, compute exact start/end page:line boundaries, preserve printed provenance, and produce a searchable, attorney-facing index.



2,027

Transcript records processed



51

Provenance-preserving chunks



57

Validated topic entries

Technical Architecture – Deterministic, Provenance-First

End-to-end pipeline converts raw PDFs into a validated topic index. Every step preserves exact printed page and line references so attorneys can verify every claim against the original transcript.



Key operational numbers: 2,027 transcript records; source range 7:11 → 88:13; 51 chunks; 5 chunks per LLM call; output treated as non-deterministic and validated deterministically.

01

Canonical Extraction

Accurate text extraction from PDF, retaining printed page and line coordinates for every record.

03

LLM Processing & Refinement

Structured topic extraction via batched LLM calls (5 chunks per call), producing candidate topics with start/end refs and evidence lists.

02

Chunking

Sequential, provenance-aligned chunks (40 records per chunk; 51 chunks here) to constrain context windows and preserve traceability.

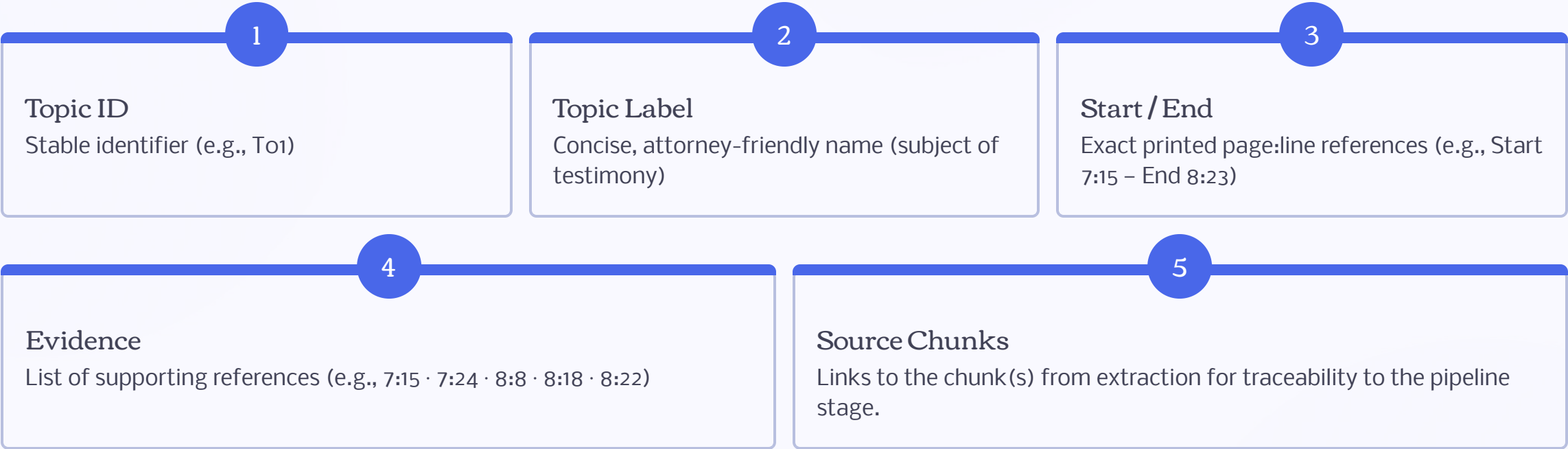
04

Deterministic Validation

Independent validator checks that every reference exists, start_ref precedes end_ref, and evidence falls within bounds – non-model logic ensures provenance integrity.

Provenance Schema – Every Topic Traces Back

Each topic entry is a compact, verifiable record that provides attorneys with immediate evidence and navigation to the original transcript.

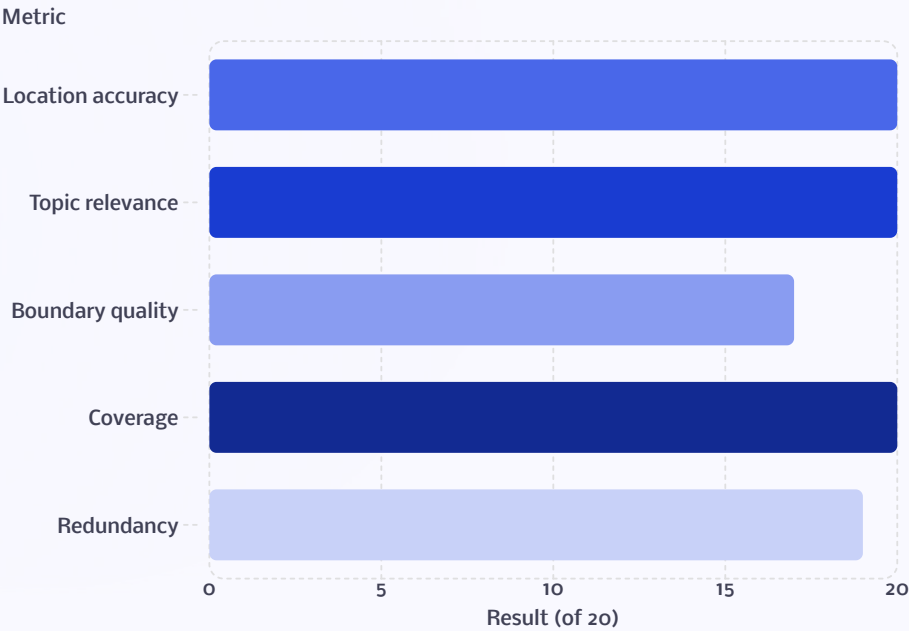


Example entry: To1 – "Deposition ground rules and admonitions" | Start 7:15 | End 8:23 | Evidence: 7:15 · 7:24 · 8:8 · 8:18 · 8:22

☐ Validation flags a plausibly named topic with incorrect source location as a failure – the system enforces exact provenance.

Current snapshot: 57 topics checked · 0 provenance errors.

Validation & Stability – Empirical Results



The system was evaluated manually across the first 20 topics and with three independent runs on the same deposition. Validation is deterministic and independent of the LLM output.

- **Three-run stability**
Run matches: Run1 vs 2: 8; Run1 vs 3: 3; Run2 vs 3: 4 – differences reflect granularity, not provenance errors.
- **Boundary references**
o invalid boundary references across all three runs – every cited page:line existed within its claimed range.
- **Key finding**
LLM segmentation varied mainly in how topics were split or merged; provenance addressing remained correct in every run.

 For litigation teams: deterministic validation creates an auditable trail that supports defensible use of AI-derived indexes during review and briefing.

Failure Analysis, Limitations & Roadmap

Practical limitations are primarily about semantic segmentation consistency (granularity and boundary drift), not provenance. These are actionable engineering and research targets.



Observed failure modes

Related topics sometimes merged or split across runs (e.g., Portfolio Transfers / Vervent / PEAKS Loans / Cancellation Rights). Boundaries drifted by several lines but always referenced valid locations.



Current limitations

Topic segmentation varies between runs; partial global re-entry detection; manual review currently covers 20 of 57 topics; validation limited to single-deposition scope so far.



Next steps

- Implement systematic boundary scoring to quantify drift.
- Improve cross-batch re-entry detection for multi-deposition threads.
- Add hierarchical parent/child topic modeling to reduce fragmentation.
- Deploy integrated Streamlit app for search, evidence review, and transcript navigation.

Resources: GitHub: github.com/Deepali-Kumari9/depoindex · Demo: to be added after deployment.

- ❑ Recommended for e-discovery teams: pilot on a small set of depositions, pair automated index with targeted manual review, and iterate using boundary scoring metrics.